MODERN ACADEMY FOR ENGINEERING & TECHNOLOGY

Electronic Eng. and Communication Tech. Branch

DSP LAB Report – Fall 27 – Law 20

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Lab-REPORT # 2

Question:

Write a MATLAB program to generate a triangle pulse with duration 2ms using 1024 samples.

1. Plot the signal in time domain using appropriate axis values.
2. Obtain and plot the spectrum of this signal.

Determine the sampling frequency:

Pulse duration = 2 ms , N = 1024 samples
Time axis from -10 ms to 10 ms (time in ms)
Ts = (10 - -10)) / N = 20 / 1024 = 0.0195 ms
Fs = 1 / Ts = 1024 / 20 = 51.2 kHz

Code:

```
N = 1024 ;
Ts = (10 -- 10) / N;
Fs = 1 / Ts;
nts = linspace(-10,10,N);
signal = tripuls( nts , 2 );
figure(1)
plot(nts,signal)
signalff = fftshift(fft(signal));
F = linspace ( -Fs/2 , Fs/2 , N);
figure(2)
plot(F , abs(signalff))
figure(3)
plot(F , angle(signalff))
```

Output:

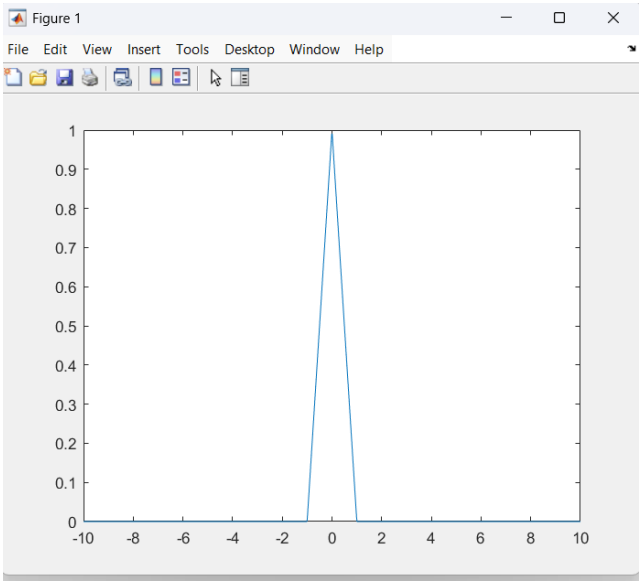


Figure (1): Triangle pulse in time domain (duration = 2 ms, time in ms)

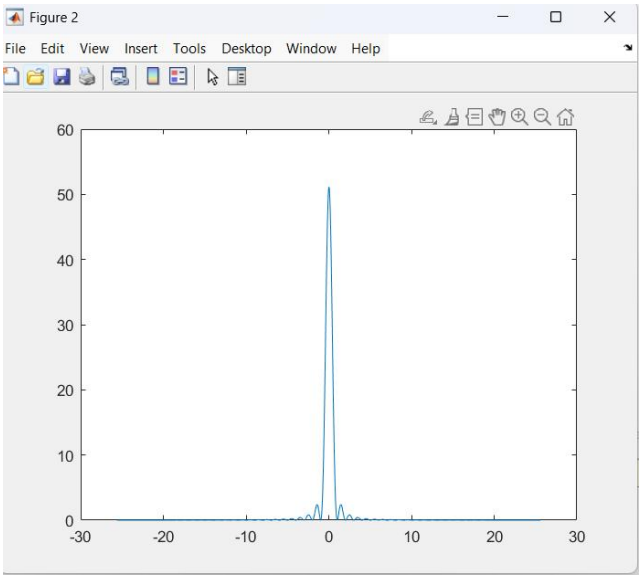


Figure (2): Amplitude spectrum of the signal |X(f)| (frequency in kHz)

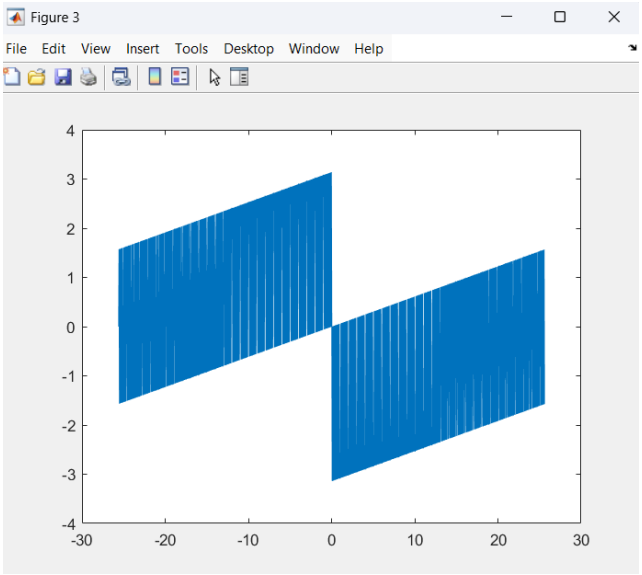


Figure (3): Phase spectrum of the signal (frequency in kHz)